

Theorem of Residual Symmetry in Primorial Quadratic Domains

A Geometric–Modular Reinterpretation of the Sieve of Eratosthenes

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Abstract

We introduce a geometric–modular formulation of coprimality with respect to primorial numbers. We show that the reflection map $y \mapsto S_n - y$, where $S_n = (P_n\#)^2$, preserves coprimality with the primorial $P_n\#$. The resulting structure induces a natural symmetry on the invertible residues modulo $P_n\#$.

We subsequently reinterpret the classical sieve of Eratosthenes as a superposition of reflective modular systems. The present work does not introduce new asymptotic estimates concerning the distribution of prime numbers; rather, it proposes a structural and geometric description of the eliminative dynamics underlying the sieve.

1. Introduction

The multiplicative group of invertible residues modulo a composite integer possesses natural symmetries related to negation modulo n . In the case of primorial numbers, these symmetries acquire a particularly regular form within the quadratic domain $S_n = (P_n\#)^2$.

The aim of this work is to formalize this reflective structure and reinterpret the sieve of Eratosthenes as a superposition of symmetric modular systems. The approach is primarily structural and expository.

2. Definitions

Definition 2.1 — Primorial

$$P_n\# = \prod_{k=1}^n p_k$$

where p_k denotes the k -th prime number.

Definition 2.2 — Primorial Quadratic Domain

$$S_n = (P_n\#)^2.$$

Definition 2.3 — Symmetric Pair

A pair (x,y) is said to be symmetric with respect to S_n if

$$x + y = S_n.$$

3. Theorem of Residual Symmetry

Theorem 3.1

Let $1 \leq y < S_n$ such that $\gcd(y, P_{n\#})=1$.

Then $x = S_n - y$ also satisfies $\gcd(x, P_{n\#})=1$.

Proof

Since every prime $p_i \leq p_n$ divides $P_{n\#}$, it also divides S_n . Hence:

$$S_n - y \equiv -y \pmod{p_i}.$$

If p_i does not divide y , then p_i does not divide $-y$. Therefore no prime $p_i \leq p_n$ divides $S_n - y$.

Consequently:

$$\gcd(S_n - y, P_{n\#})=1.$$

4. Modular Interpretation

The set of integers coprime with $P_{n\#}$ coincides with the multiplicative group

$$(\mathbb{Z} / P_{n\#} \mathbb{Z})^\times.$$

The transformation $y \mapsto -y$ is an involutive automorphism of this group.

The reflection $y \mapsto S_n - y$ realizes this negation in the integer domain because

$$S_n \equiv 0 \pmod{P_{n\#}}.$$

5. Arithmetic Consequences

Corollary 5.1

Let $1 < y \leq P_{n\#}$ such that $\gcd(y, P_{n\#})=1$.

Then every prime factor of y is greater than p_n .

Corollary 5.2

Let $1 < y < p_{n+1}^2$ and suppose $\gcd(y, P_{n\#}) = 1$.

Then y is prime.

Proof

If y were composite, it would possess a prime factor $q \leq \sqrt{y} < p_{n+1}$.

Hence $q \leq p_n$, implying that q divides $P_{n\#}$, contradicting $\gcd(y, P_{n\#}) = 1$.

Therefore y must be prime.

6. Complete Reflective Elimination

Let $N \in \mathbb{N}$ and let S be a primorial quadratic center.

For every prime $p \leq \sqrt{N}$, define the modular reflective center

$$C_p = S - (S \bmod p).$$

Then $C_p \equiv 0 \pmod{p}$.

The reflection $x \mapsto C_p - x$ preserves divisibility modulo p .

Indeed, if $x \equiv 0 \pmod{p}$, then also $C_p - x \equiv 0 \pmod{p}$.

It follows that the set of multiples of p ,

$$E_p = \{x : x \equiv 0 \pmod{p}\},$$

is symmetric with respect to the center C_p .

The superposition of the eliminating families

$$\bigcup_{p \leq \sqrt{N}} E_p$$

coincides with the set of composite numbers in the interval $[2, N]$.

The complement of this superposition consists exclusively of the prime numbers and the integer 1.

7. Geometric Interpretation of the Sieve

The classical sieve of Eratosthenes may therefore be reinterpreted as a superposition of reflective modular systems.

Each prime modulus generates:

- a dynamic reflective center;
- a modular periodicity;
- a symmetric elimination family;
- a geometric decomposition of the arithmetic domain.

Within this perspective, prime numbers emerge as points not eliminated by any reflective modular system.

8. Discussion

The results presented derive from classical properties of coprimality and multiplicative structures modulo n .

The principal contribution of the present work lies in the geometric and reflective reinterpretation of these properties in the context of primorial numbers.

No new asymptotic estimates on the distribution of prime numbers are introduced, nor does the framework provide computational improvements over classical sieve methods. The interest of the proposed approach is therefore primarily structural and conceptual.

9. Conclusion

The reflection $y \mapsto S_n - y$ preserves coprimality with respect to primorial numbers and induces a natural symmetric structure on invertible residues.

This observation allows a geometric reinterpretation of the sieve of Eratosthenes as a superposition of reflective modular systems.

The framework suggests possible future connections between residue theory, discrete geometry, and periodic modular structures.

References

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